Mian Saad Karim

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PROFILE

Hardware and Embedded Systems Engineer with hands-on experience in PCB design, custom flight-controller firmware, edge AI deployment, and UAV swarm research. Proficient across the full product lifecycle—from schematic capture and 4-layer PCB layout to firmware bring-up (ESP-IDF, ArduPilot) and field-tested drone assembly.

EDUCATION

Pakistan Institute of Engineering & Applied Sciences (PIEAS) Graduated Jul 2026
BS in Electrical Engineering (CGPA: 3.64 / 4.00) *USAID Merit Scholar (3-Year Award, 180k PKR/sem)*

EXPERIENCE

Embedded Systems Engineer, *QuickGen Technologies* Aug 2026 – Present

- Developing end-to-end client embedded solutions; designed AI-enabled calculator (STM32 + Raspberry Pi CM5 in calculator footprint) and screenless NFC smart speaker for children (ESP32-S3 + SD card audio).

Hardware Design & Verification Engineer, *Signatics* Jul 2026 – Present

- Designing high-power multi-rail DC power distribution systems, TVS surge protection, and automated test fixtures to validate commercial products and SDR platforms.

Electrical Engineer (Part-time & Intern), *Saf Shikan* Jul 2025 – Jun 2026

- Prototyped ESP32-based autonomous mini-drones; engineered custom flight-controller firmware, optical-flow sensor integration, and motor noise filtering.

Intern – Quest Program, *NASTP* Jul 2025 – Aug 2025 · **Avionics Intern**, *PAC Kamra* Jul 2024

- Developed STEM educational kits (secured NASTP budget); gained avionics production and RF screening experience at PAC.

PROJECTS

Thesis: UAV Swarm for Autonomous Disaster Assessment GitHub

- Sole hardware/firmware lead in a 3-person team at PIEAS; personally owned 100% of the 4-layer avionics PCB design, DMA camera drivers, and dual-MCU system architecture.
- Engineered 4-layer PCB isolating STM32F405 (ArduPilot flight loop) from ESP32-S3 (Edge AI vision, MAVLink v2 bridge, and ad-hoc ESP-NOW mesh telemetry); board doubles as structural airframe.
- **Hard Problem Solved:** Diagnosed severe high-current motor inductive switching noise causing 3.3V brownouts and IMU yaw drift; eliminated failures by redesigning to star grounding, localized LC filtering, and flyback diode suppression.
- **Autonomous Swarm Outcome:** Flight-tested across 2 physical drones + 1 ground bridge node communicating over decentralized ESP-NOW mesh with autonomous waypoint navigation and real-time survivor detection.

ESP32-S3 Fire Detection + Pixhawk Companion Computer GitHub

- Combined Edge Impulse FOMO fire detection with a MAVLink v2 companion computer on an ESP32-S3.
- Web dashboard: live camera feed with AI overlay, arm/disarm, RC override, auto-flight, and full telemetry.

Flywing Mini Clone: 3-PCB Modular Flight Stack Fiverr Freelance

- Ground-up PCB clone of SpeedyBee F405 Wing Mini: FC (STM32F405, ICM-42688-P, AT7456E OSD), PDB, USB-C breakout.

FPGA-Based Dual-Axis Servo Control with UART + GUI GitHub

- Real-time servo control in Verilog on Spartan-3E FPGA with Python/Tkinter GUI over UART.

VGA Graphics & Game on Spartan-3E FPGA GitHub

- Minimalist airplane game in Verilog generating 640×480 VGA signal entirely through RTL.

TECHNICAL SKILLS

PCB Design: Altium Designer, EasyEDA (schematic + layout), Gerber generation, JLCPCB; 2-layer and 4-layer stackups
MCUs & Firmware: ESP32, ESP32-S3, STM32F405; ESP-IDF, ArduPilot, INAV
Protocols: MAVLink v2, ESP-NOW, SPI, I^2C, UART, USB-C
FPGA / HDL: Verilog on Xilinx Spartan-3E (VGA, UART, PWM generation)
Edge AI: Edge Impulse (FOMO object detection), TinyML on ESP32-S3
Languages & Tools: C, C++, Python, Verilog, MATLAB, Git/GitHub, Linux CLI, Onshape CAD

HONORS & AWARDS

USAID Merit Scholarship – USAID & HEC (3-Year Award, 180k PKR/semester) 2023 – 2026
ISYWSC'24 Runner-up, Electro Masters – IEEE Dec 2024